



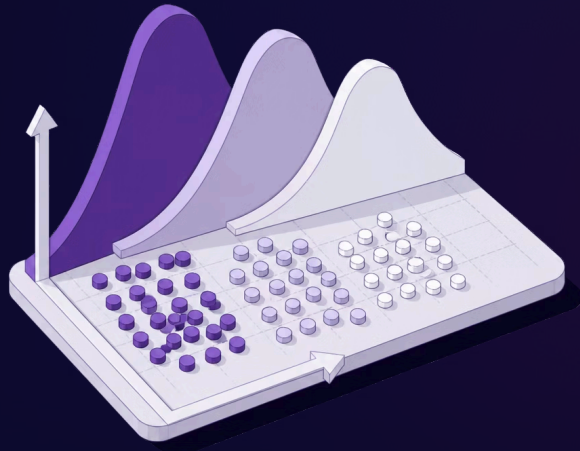
Gaussian Mixture Models (GMM)

A probabilistic approach to clustering in Machine Learning

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Machine learning

What is a Gaussian Mixture Model?



Gaussian Distribution

A bell-shaped curve describing how data spreads around a mean

Multiple Gaussians Mixed

GMM assumes data is generated from **several** Gaussian distributions blended together

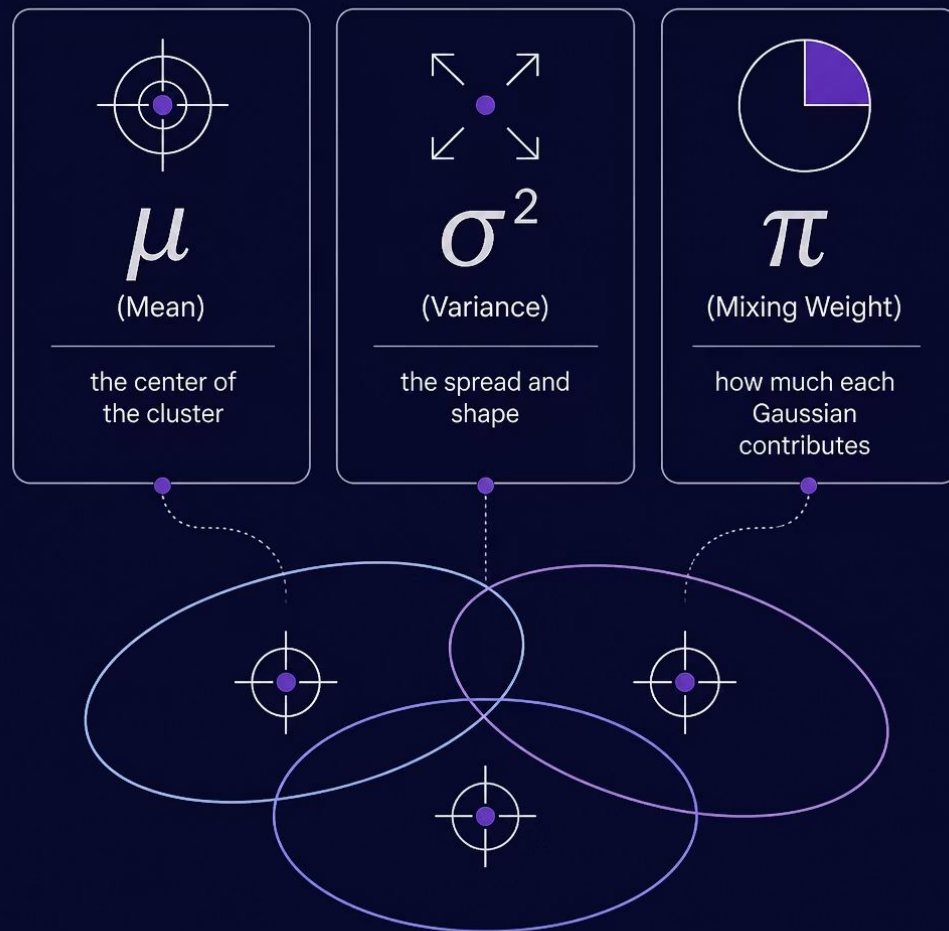
Each Component = A Cluster

Each Gaussian has its own shape, size, and center

Probabilistic Assignment

Every data point gets a **probability score** for each cluster — not a hard label

How GMM Works: The Three Key Parameters



Mean (μ)

The **center** of each Gaussian cluster — where the distribution peaks



Variance / Covariance (σ^2)

Controls the **spread and shape** of each cluster — circular or elliptical

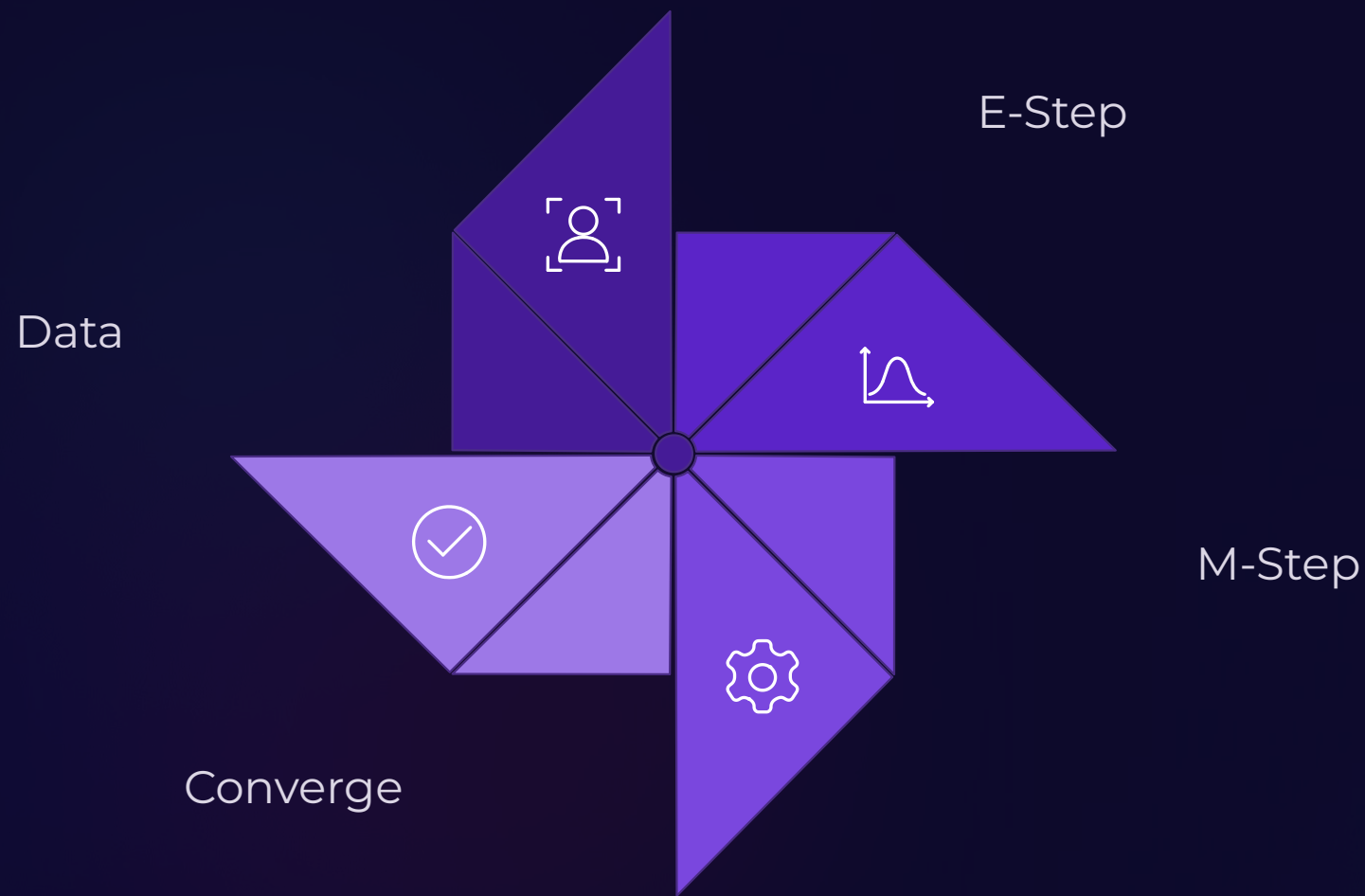


Mixing Weight (π)

How much **each Gaussian contributes** to the overall model

- Each data point receives a **soft probability score** for every cluster — never a hard single label.

The Expectation-Maximization (EM) Algorithm



Data

Load the dataset



E-Step

Assign probabilities per Gaussian



M-Step

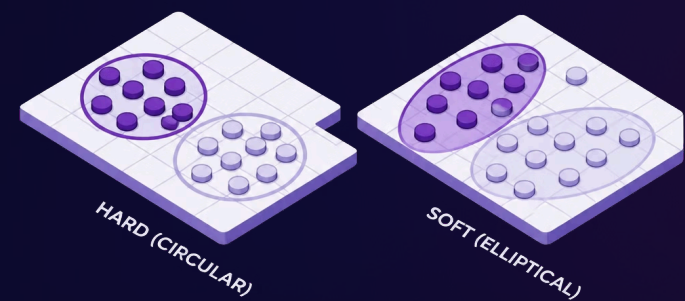
Update μ , σ^2 , and π



Convergence

Stop when parameters stabilize

GMM vs K-Means: A Clear Comparison



Feature	K-Means	GMM
Clustering Type	Hard (one cluster only)	Soft (probabilities)
Assignment Method	Distance-based	Probability-based
Cluster Shape	Circular / Spherical	Flexible (elliptical)
Speed	Faster	Slower
Overlapping Data	Poor accuracy	High accuracy

Advantages, Applications & Summary

KEY STRENGTHS

✓ Soft Clustering

Probabilities, not hard labels

✓ Overlapping Clusters

Handles them naturally

✓ Flexible Shapes

Elliptical, not just circular

✓ Uncertainty Quantification

Real probability estimates

REAL-WORLD APPLICATIONS

→  Image Segmentation

→  Customer Segmentation

→  Speech Recognition

→  Anomaly Detection

→  Medical Data Analysis

📄 **Summary:** GMMs represent data as a combination of multiple Gaussian distributions and assign **probabilities** instead of hard cluster labels.